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EXPERIMENT 3

PERFORM SESSION HIJACKING USING VARIOUS TOOLS

AIM:

To understand and demonstrate session hijacking in an authorized lab environment using Ettercap and Wireshark.

TOOLS REQUIRED:

1. **Kali Linux:** Used as the monitoring and testing machine.
2. **Metasploitable2:** Vulnerable target machine hosting the DVWA web application.
3. **VirtualBox:** Used to run Kali Linux and Metasploitable2 in an isolated lab network.
4. **Wireshark:** Used to capture and analyse network traffic.
5. **Mozilla Firefox:** Used to access DVWA and inspect browser storage.
6. **DVWA (Damn Vulnerable Web Application):** Used as the intentionally vulnerable web application for session security testing.
7. **Terminal utilities:** ip a, ifconfig, and ping to identify IP addresses and verify connectivity.

PROCEDURE:

1. Start Kali Linux and Metasploitable2 in VirtualBox and ensure both machines are connected to the same isolated host-only network.
2. Identify the IP addresses of both machines using the following commands:

   Kali Linux: ip a

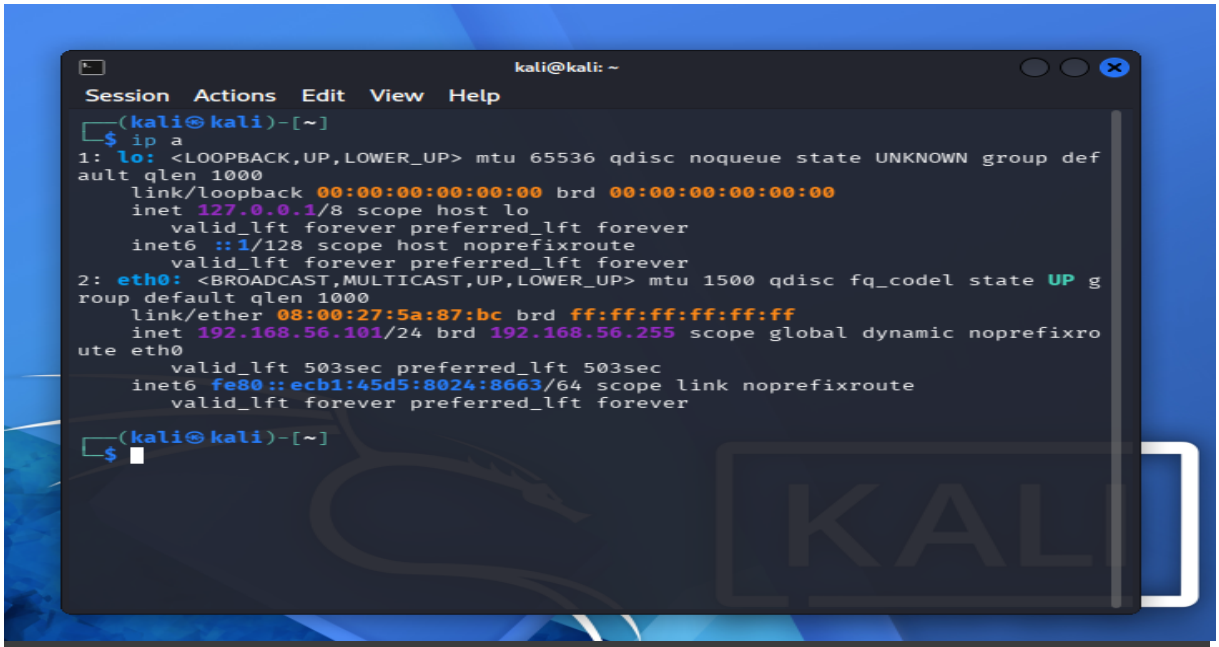
   Metasploitable2: ifconfig
3. Verify connectivity between the machines using the ping command:

   ping -c 4 192.168.56.102
4. Open Wireshark in Kali Linux and start capturing packets on the eth0 network interface.

5. Apply the display filter http to observe HTTP requests and responses generated while accessing the target web server.
6. Open the DVWA application hosted on Metasploitable2 using Firefox and log in to the application.
7. Open Firefox Developer Tools and navigate to Storage → Cookies to inspect the DVWA session cookie and its security attributes, without exposing or reusing the session token.
8. Record the observed network traffic, session cookie attributes, and security implications of transmitting session information over unencrypted HTTP.
9. Analyse the observations and document the security risks associated with session hijacking and possible preventive measures.

OBSERVATION:

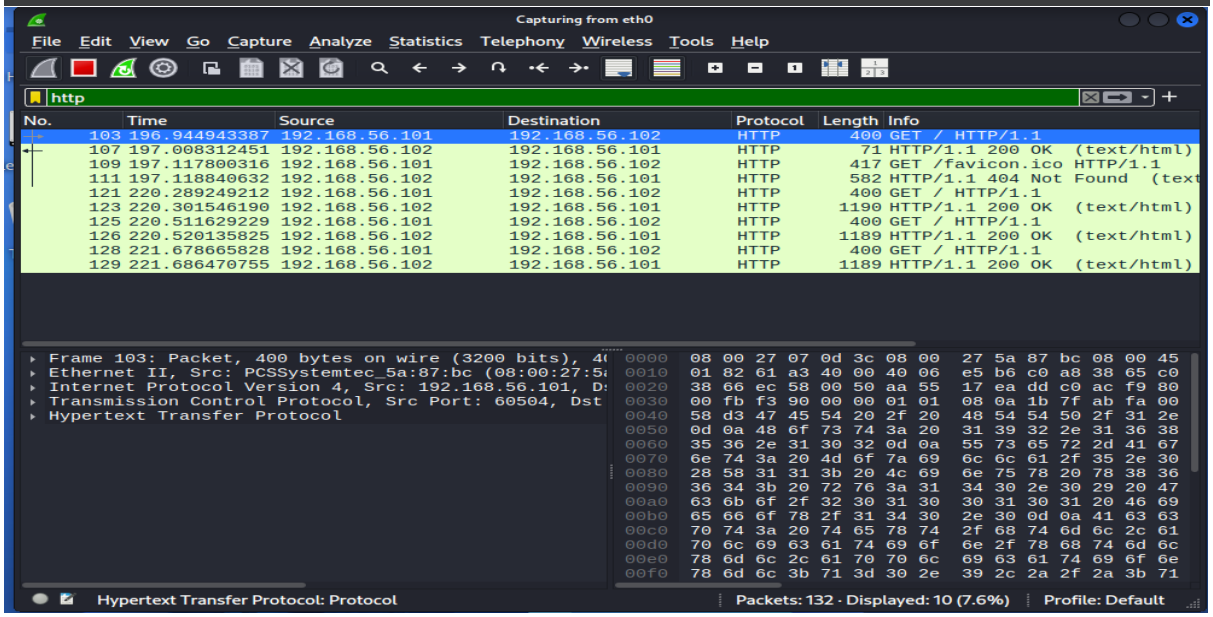
OBSERVATION	**RESULT**
Kali Linux IP address	192.168.56.101
Metaspoitable2 IP address	192.168.56.102
Network connectivity	Successful; 4 packets received with 0% packet loss
Wireshark packet capture	HTTP GET requests and HTTP 200 OK responses observed
Test application	DVWA opened successfully; loged in as admin
Session cookie	PHPSESSID observed in Firefox Storage
Cookie security attributes	HttpOnly: false; Secure: false; Samesite: None, displayed in the captured screenshot
Session hijacking	No

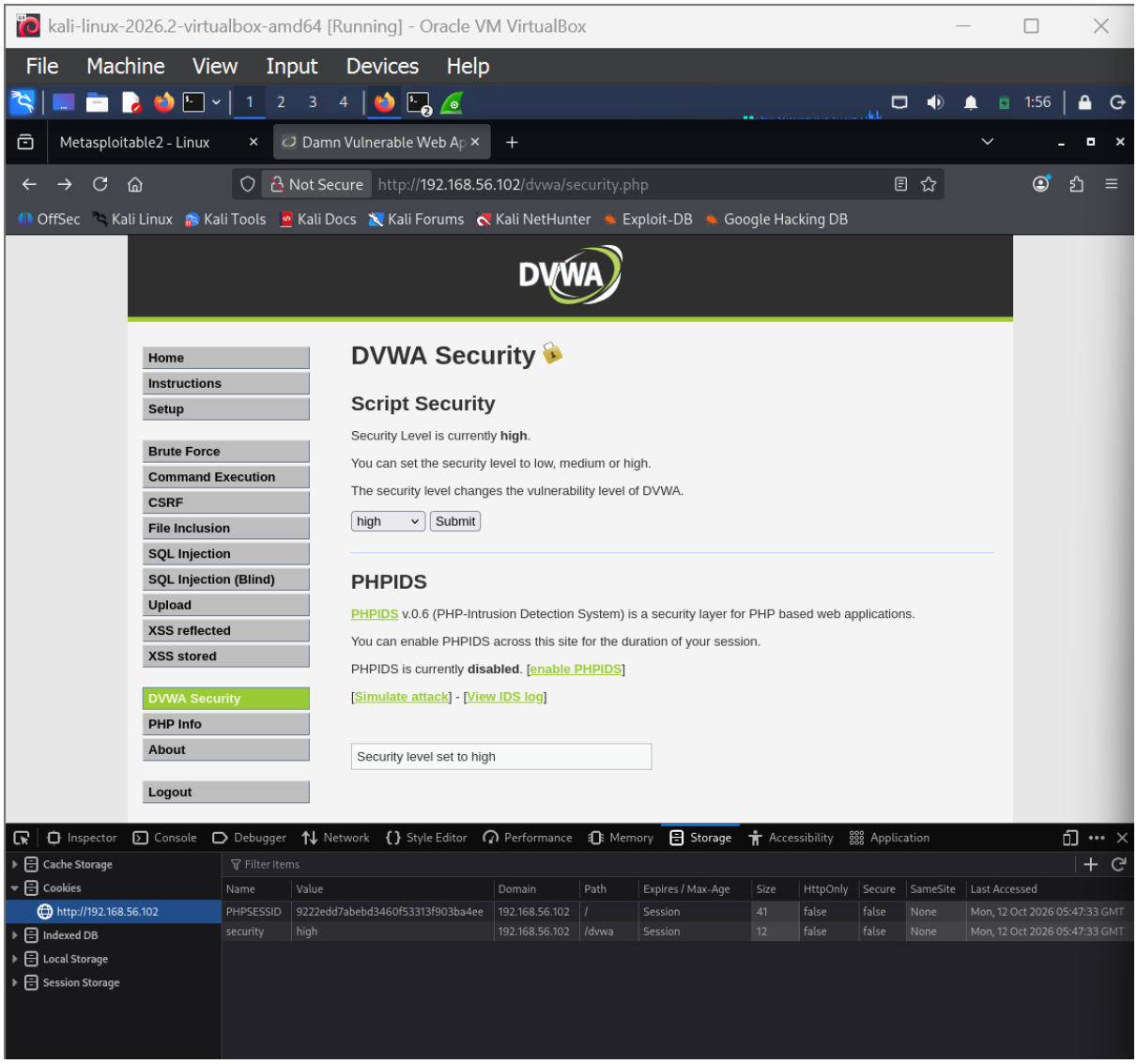



```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:07:0d:3c
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe07:d3c/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1188 (1.1 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:96 errors:0 dropped:0 overruns:0 frame:0
          TX packets:96 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21437 (20.9 KB)  TX bytes:21437 (20.9 KB)

msfadmin@metasploitable:~$
```





RESULT:

HTTP traffic was captured and analysed using Wireshark. The DVWA session cookie and its security attributes were inspected using Firefox Developer Tools, and the security risks associated with session hijacking were studied.